CONSTRUCTIBILITY OF R^QF_* OF A TORSION FINITE MODULE ALONG A PROPER SMOOTH FAMILY OF CURVES WITH GEOMETRICALLY CONNECTED FIBRES

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Allowed standard prerequisites (editorial).

- Constructible abelian sheaves closed under extensions (Etale Cohomology, Lemma etale-cohomology-lemma-constructible-abelian, canonical tag 03RZ). cited at proof lines 4604-4606. Runs the induction on the torsion degree: the long exact sequence sandwiches R^qf_*M between constructible terms once the prime case is known.
- Projection formula for proper pushforward of mod-n sheaves (Etale Cohomology, Lemma etale-cohomology-lemma-projection-formula-proper-mod-n, canonical tag 0F0G). cited at proof lines 4649-4656. Trades Rf_*Lambda for the derived tensor of Rf_*Z/lZ with Lambda, connecting the free-module case to the constant sheaf case.
- Constant sheaf case (Lemma lemma-proper-smooth-family-curves, canonical tag 0GKV). cited at proof line 4667. The terminal base case: constructibility of R^qf_*Z/lZ for the proper smooth family of curves.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Cited at proof lines 4642-4644: the first spectral sequence reduces constructibility of the cohomology sheaves of the truncated free complex to the free terms.

Source reference. Original lemma 07K7: sources/stacks-chow/tags.txt, source lines 9647–9647. Cited at proof lines 4629-4634: with the finite cohomological dimension lemma it truncates the free resolution to a bounded complex.

Source reference. Original lemma 095U: sources/stacks-chow/tags.txt, source lines 11597–11597. Cited at proof lines 4629-4632: f_* has finite cohomological dimension on sheaves of Lambda-modules, enabling the truncation.

Source reference. Original lemma 03RZ: sources/stacks-chow/tags.txt, source lines 4948–4948. Cited at proof lines 4604-4606: the outer terms of the long exact sequence are constructible, winning the induction on n.

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Source reference. Original lemma 0F0G: sources/stacks-chow/tags.txt, source lines 18761–18761. Cited at proof lines 4649-4656: (Rf_*Z/lZ) tensor^L Lambda = Rf_*Lambda, the key trade to the constant sheaf case.

Source reference. Original lemma 0GKB: sources/stacks-chow/tags.txt, source lines 20681–20681. Cited at proof lines 4664-4666: tensoring the constructible R^qf_*Z/lZ with Lambda stays constructible.

Source reference. Original lemma 0GKV: sources/stacks-chow/tags.txt, source lines 20700–20700. Cited at proof line 4667: the terminal base case, constructibility of R^qf_*Z/lZ.

Source reference. Original lemma 0D6J: sources/stacks-chow/tags.txt, source lines 16524–16524. Cited at proof lines 4611-4616: R^qf_*M does not change when M is viewed as a sheaf of Lambda/lLambda-modules.

Lemma 0.2. *Let $f : X \to S$ be a proper smooth morphism of schemes with geometrically connected fibres of dimension 1. Let Λ be a Noetherian ring. Let M be a finite Λ-module annihilated by an integer $n > 0$. Then $R^q f_* \underline{M}$ is a constructible sheaf of Λ-modules on S.*

Proof. If $n = \ell n'$ for some prime number ℓ, then we get a short exact sequence $0 \to M[\ell] \to M \to M' \to 0$ of finite Λ-modules and M' is annihilated by n'. This produces a corresponding short exact sequence of constant sheaves, which in turn gives rise to an exact sequence

$$R^{q-1} f_* \underline{M'} \to R^q f_* \underline{M[n]} \to R^q f_* \underline{M} \to R^q f_* \underline{M'} \to R^{q+1} f_* \underline{M[n]}$$

Thus, if we can show the result in case M is annihilated by a prime number, then by induction on n we win by Étale Cohomology, Lemma 03RZ.

Let ℓ be a prime number such that ℓ annihilates M. Then we can replace Λ by the $\mathbf{F}_\ell$-algebra $\Lambda/\ell\Lambda$. Namely, the sheaf $R^q f_* \underline{M}$ where $\underline{M}$ is viewed as a sheaf of Λ-modules is the same as the sheaf $R^q f_* \underline{M}$ computed by viewing $\underline{M}$ as a sheaf of $\Lambda/\ell\Lambda$-modules, see Cohomology on Sites, Lemma 0D6J.

Assume ℓ be a prime number such that ℓ annihilates M and Λ. Let us reduce to the case where M is a finite free Λ-module. Namely, choose a resolution

$$\ldots \to \Lambda^{\oplus m_2} \to \Lambda^{\oplus m_1} \to \Lambda^{\oplus m_0} \to M \to 0$$

Recall that f_* has finite cohomological dimension on sheaves of Λ-modules, see Étale Cohomology, Lemma 095U and Derived Categories, Lemma 07K7. Thus we see that $R^q f_* \underline{M}$ is the qth cohomology sheaf of the object

$$Rf_*(\underline{\Lambda^{\oplus m_a}} \to \ldots \to \underline{\Lambda^{\oplus m_0}})$$

in $D(S_{étale}, \Lambda)$ for some integer a large enough. Using the first spectral sequence of Derived Categories, Lemma 015J (or alternatively using an argument with truncations) we conclude that it suffices to prove that $R^q f_* \underline{\Lambda})$ is constructible.

At this point we can finally use that

$$(Rf_* \underline{\mathbf{Z}/\ell\mathbf{Z}}) \otimes^{\mathbf{L}}_{\mathbf{Z}/\ell\mathbf{Z}} \underline{\Lambda} = Rf_* \underline{\Lambda}$$

by Étale Cohomology, Lemma 0F0G. Since any module over the field $\mathbf{Z}/\ell\mathbf{Z}$ is flat we obtain

$$(R^q f_* \underline{\mathbf{Z}/\ell\mathbf{Z}}) \otimes_{\mathbf{Z}/\ell\mathbf{Z}} \underline{\Lambda} = R^q f_* \underline{\Lambda}$$

Hence it suffices to prove the result for $R^q f_* \underline{\mathbf{Z}/\ell\mathbf{Z}}$ by Étale Cohomology, Lemma 0GKB. This case is Lemma 0GKV. □